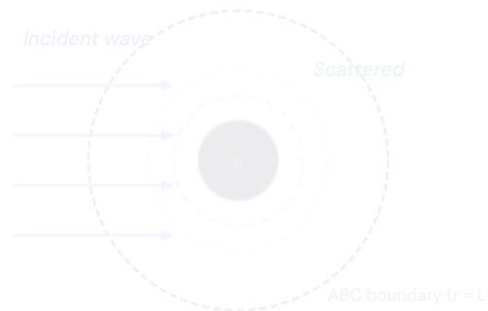


Physics-Informed Neural Network for 2D Acoustic Scattering

Solving the Helmholtz equation across wavenumbers $ka = 0.5$ to 2π with 2–8% L2 error, under 3% for $ka \geq \pi$

Carl Merrigan — March 2026



Problem: Acoustic Wave Scattering

A plane wave impinges on a rigid circular cylinder. We predict the **scattered field** — the perturbation caused by the obstacle.

$$\nabla^2 \phi_s + k^2 \phi_s = 0$$

The dimensionless parameter **ka** (wavenumber × radius) controls difficulty: higher ka = more oscillatory field = finer spatial features.

We train from ka = 0.5 (sub-wavelength) through ka = 2π (2 wavelengths per radius), spanning a 12× range of spatial frequency.

Exact Solution (Validation)

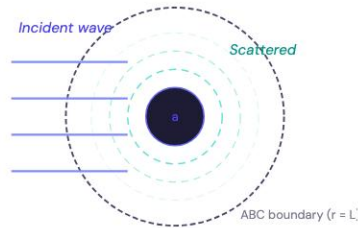
The Jacobi-Anger identity expands the incident wave in cylindrical harmonics:

$$\phi_{\text{inc}} = e^{ikr \cos \theta} = \sum_{n=-\infty}^{\infty} i^n J_n(kr) e^{in\theta}$$

The sound-hard Neumann condition $\partial \phi_{\text{total}} / \partial r|_{r=a} = 0$ yields the exact scattered field:

$$\phi_s = - \sum_{n=-\infty}^{\infty} i^n \frac{J'_n(ka)}{H_n^{(1)'}(ka)} H_n^{(1)}(kr) e^{in\theta}$$

Converges for $N \approx ka + 20$ terms. This exact solution provides pixel-level validation of the PINN — no numerical reference solver needed.



Scattered-Field Formulation

The network outputs $(u, v) = (\text{Re}(\phi_s), \text{Im}(\phi_s))$. The total field is reconstructed as $\phi_{\text{total}} = e^{ikx} + \phi_s$. Three loss terms enforce physics:

PDE RESIDUAL

$$\nabla^2 \phi_s + k^2 \phi_s = 0$$

NEUMANN BC ($r = a$)

$$\frac{\partial \phi_s}{\partial n} = -\frac{\partial \phi_{\text{inc}}}{\partial n}$$

BGT2 ABC ($r = L$)

$$\frac{\partial \phi_s}{\partial r} - ik\phi_s + \frac{\phi_s}{2r} = 0$$

TOTAL LOSS

$$\mathcal{L} = \lambda_{\text{pde}} \mathcal{L}_{\text{pde}} + \lambda_{\text{bc}} \mathcal{L}_{\text{bc}} + \lambda_{\text{abc}} \mathcal{L}_{\text{abc}}$$

Scattered-field formulation lets the network focus on the unknown perturbation, avoiding the need to learn the known incident plane wave.

Network Architecture



Random Fourier Features

$$\gamma(\mathbf{x}) = [\sin(B\mathbf{x}), \cos(B\mathbf{x})]$$

Overcomes spectral bias — standard MLPs struggle with high-frequency functions in low-dimensional domains [Tancik et al, NeurIPS 2020]

Bandwidth σ set to wavenumber k , encoding the physical length scale directly into the feature space.

Design Choices

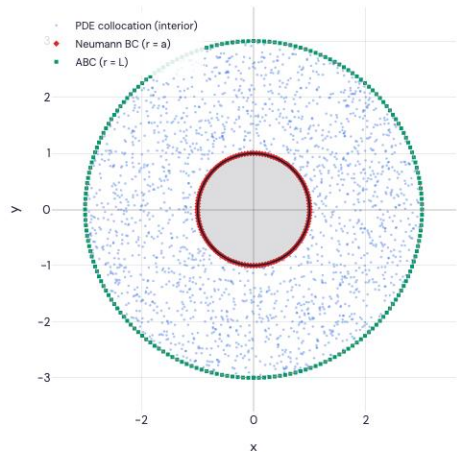
- **Residual blocks:** skip connections stabilize deep training
- **Xavier init:** balanced gradient flow at initialization
- **Tanh activation:** smooth, matches solution regularity
- **Complex output:** (u, v) avoids branch cuts in polar form

~296K parameters (FF-PINN) vs ~298K (plain MLP baseline)

Training Strategy & Loss Functions

Collocation Sampling

Training Collocation Points



Two-Phase Optimization

- Phase 1 — Adam** (10K–50K epochs)
Cosine annealing LR; resampling points every 2K epochs
- Phase 2 — L-BFGS** (200–300 iters)
Strong Wolfe line search; fixed points for stable Hessian

High-ka: wider network (384–512n), more Fourier features (96–128), 50K Adam epochs.

Loss Components

COMPONENT	DEFINITION	Λ
$\mathcal{L}_{\text{pde}}$	$\frac{1}{N} \sum \nabla^2 \phi_s + k^2 \phi_s ^2$	1.0
$\mathcal{L}_{\text{bc}}$	$\frac{1}{N} \sum \partial_n \phi_s + \partial_n \phi_{\text{inc}} ^2$	10.0
$\mathcal{L}_{\text{abc}}$	$\frac{1}{N} \sum \partial_r \phi_s - ik \phi_s + \phi_s/2r ^2$	1.0

Evaluation Metrics

vs analytic Bessel/Hankel series, 200×200 grid.

METRIC	DEFINITION
L2 Relative	$\ \phi_s^{\text{PINN}} - \phi_s^{\text{exact}}\ _2 / \ \phi_s^{\text{exact}}\ _2$ — primary, normalized
Max Error	$\max \phi_s^{\text{PINN}} - \phi_s^{\text{exact}} $ — worst-case
Mean Error	$\bar{e} = \frac{1}{N} \sum \phi_s^{\text{PINN}} - \phi_s^{\text{exact}} $ — typical

All errors use complex magnitude $|\phi| = \sqrt{u^2 + v^2}$.

Ablation: Fourier Features

Parameter-matched comparison — FF-PINN (64 features, 256 hidden, 296K params) vs plain MLP (272 hidden, 298K params). Identical 10K Adam + 200 L-BFGS schedules (shorter than production) to isolate architecture.

$ka = \pi$ (low frequency)

	FF-PINN	PLAIN MLP
L2 relative	2.42%	2.42%
Max error	2.97%	2.99%

Identical — spectral bias not a bottleneck at low frequency.

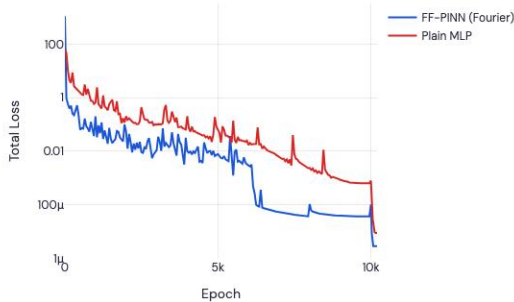
Key insight: The plain MLP's L-BFGS phase *increased* error (10.9%→11.7%), while the FF-PINN's L-BFGS *halved* it (19.6%→10.1%). Fourier features create a smoother loss landscape that second-order optimizers can exploit.

$ka = 2\pi$ (high frequency)

	FF-PINN	PLAIN MLP	RATIO
L2 relative	10.1%	11.7%	1.15×
Max error	22.7%	47.3%	2.1×

Max error doubles without Fourier features.

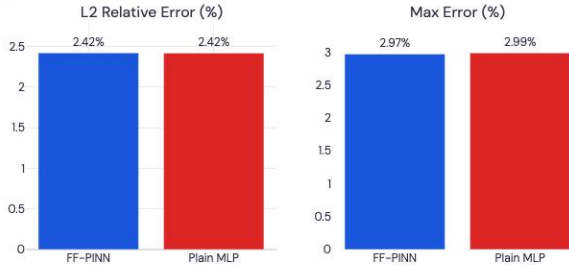
Ablation: Loss Convergence ($ka=\pi$)



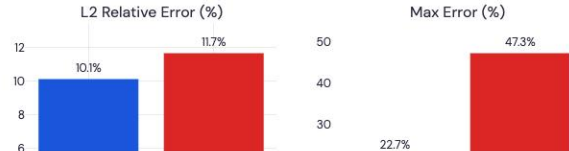
Ablation: Loss Convergence ($ka=2\pi$)



Ablation: Final Error ($ka=\pi$)



Ablation: Final Error ($ka=2\pi$)



Absorbing Boundary Conditions

First-Order ABC

$$\frac{\partial \phi_s}{\partial n} - ik\phi_s = 0$$

Assumes purely outgoing plane waves at the boundary. Works on any shape but ignores wavefront curvature.

BGT2 (Second-Order)

$$\frac{\partial \phi_s}{\partial r} - ik\phi_s + \frac{\phi_s}{2r} = 0$$

Adds a curvature correction $\phi_s/2r$ that accounts for the $1/\sqrt{r}$ amplitude decay of cylindrical waves

[Bayliss, Gunzburger & Turkel, 1982]

Why Circular Domain?

- **Symmetry match:** circular boundary matches the cylindrical geometry of the scatterer — uniform distance to the obstacle from all points
- **Well-defined normals:** square boundaries have corner singularities where the outward normal is undefined → ABC is ill-conditioned
- **BGT2 requires curvature:** the $\phi_s/2r$ correction is derived for circular boundaries; it has no square-boundary analogue

Result: Circle + BGT2 gives the most physically consistent setup for this cylindrical scattering problem.

Production Results

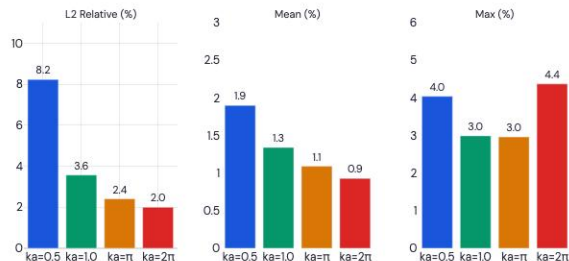
2.00 %

Best L2 error (ka=2 π)

4×

ka values validated

KA	L2 REL ERROR	MEAN ERROR	MAX ERROR	NETWORK	EPOCHS	RUNTIME
ka = 0.5	8.23%	1.90%	4.04%	256 / 4L / 64ff	10K+200	12 min
ka = 1.0	3.57%	1.34%	2.99%	256 / 4L / 64ff	10K+200	13 min
ka = π	2.41%	1.09%	2.96%	256 / 4L / 64ff	10K+200	17 min
ka = 2 π	2.00%	0.93%	4.37%	384 / 6L / 96ff	50K+300	259 min

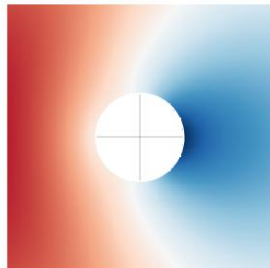


All runs: circle boundary, BGT2 ABC, L = 3.0, trained on 4× NVIDIA RTX 4000 Ada. Mean error < 2% everywhere; max error peaks at 4.4% for ka=2 π (localized to the shadow boundary).

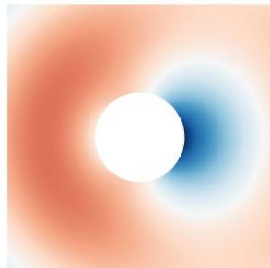
L2 relative error is highest at low ka because the scattered field amplitude is small (smaller normalization denominator) and these runs received 4× less compute (10K vs 50K epochs, smaller network). The low-ka errors could likely be reduced with comparable training budgets.

Multi-Scale Scattering Physics

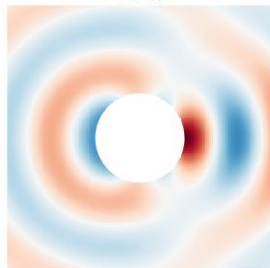
Analytic Scattered Field — Increasing Wavenumber
 $ka = 0.5$



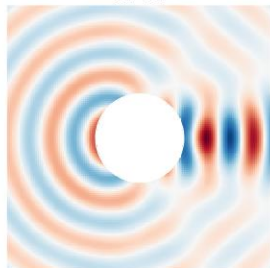
$ka = 1.0$



$ka = \pi$



$ka = 2\pi$



Analytic scattered field $\text{Re}(\Phi_s)$ across four wavenumbers, showing how the scattering pattern evolves from a simple dipole at $ka = 0.5$ to a complex multi-lobe pattern at $ka = 2\pi$.

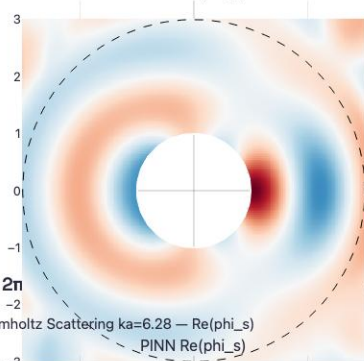
- **$ka = 0.5$:** Sub-wavelength — weak, dipolar scattering
- **$ka = 1.0$:** Transition regime — shadow begins to form
- **$ka = \pi$:** Resonance — strong diffraction, clear shadow
- **$ka = 2\pi$:** Multi-wavelength — complex interference fringes

The PINN must resolve increasingly fine spatial features as ka grows — motivating Fourier features with $\sigma = k$.

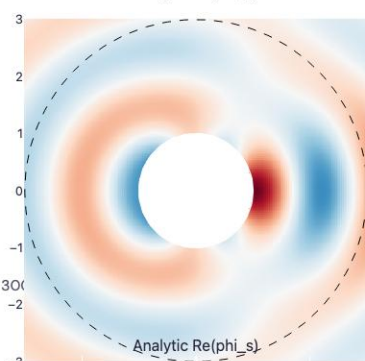
PINN vs Analytic: Field Comparisons

$ka = \pi$ **2.41** % L2 2.96% max

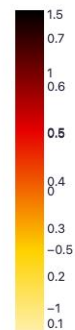
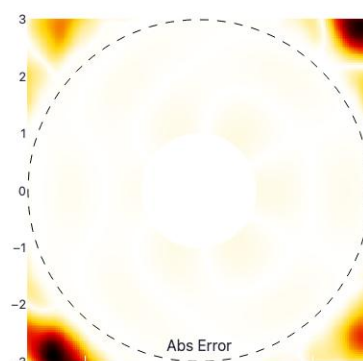
Helmholtz Scattering $ka=3.14 - \text{Re}(\phi_{i,s})$
PINN $\text{Re}(\phi_{i,s})$



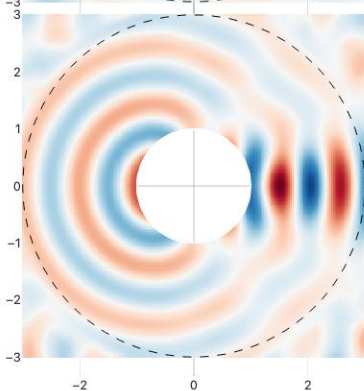
Analytic $\text{Re}(\phi_{i,s})$



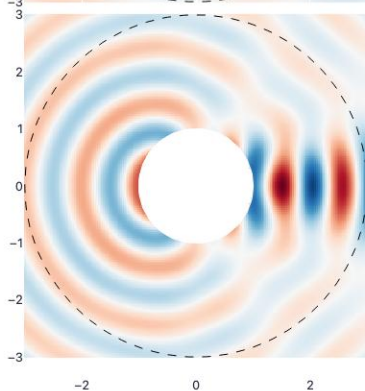
Abs Error



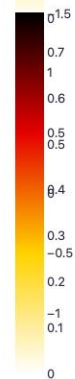
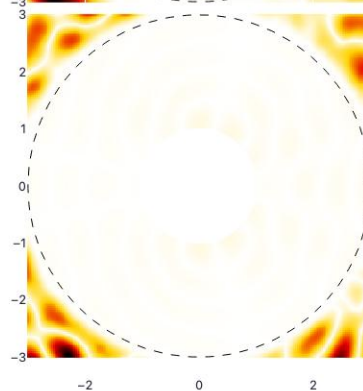
$ka = 2\pi$
Helmholtz Scattering $ka=6.28 - \text{Re}(\phi_{i,s})$
PINN $\text{Re}(\phi_{i,s})$



50K Adam + 30K
Analytic $\text{Re}(\phi_{i,s})$



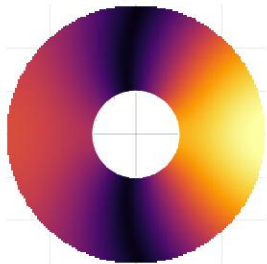
Abs Error



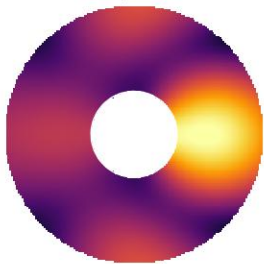
Spatial Error Distribution

Absolute Error |PINN - Analytic|

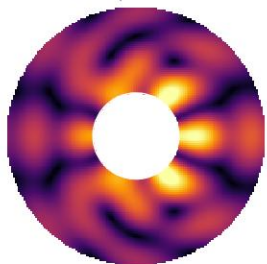
$ka = 0.5$ | $L2 = 8.23\%$



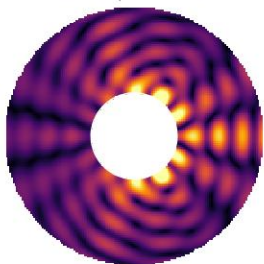
$ka = 1.0$ | $L2 = 3.57\%$



$ka = \pi$ | $L2 = 2.41\%$



$ka = 2\pi$ | $L2 = 2.00\%$

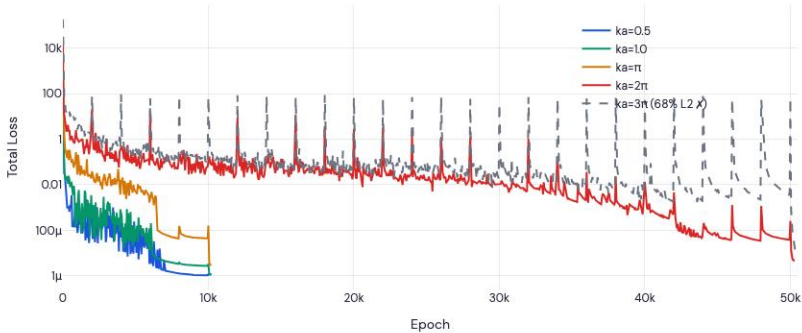


Absolute error |PINN - analytic| across all four production wavenumbers. Error is concentrated near the scatterer surface and in the shadow region behind the cylinder — the regions with the steepest field gradients.

- **$ka = 0.5, 1.0$:** errors uniformly low, smooth fields are easy for the network
- **$ka = \pi$:** shadow boundary shows slightly elevated error where the field transitions sharply
- **$ka = 2\pi$:** error peaks near the surface where the oscillation wavelength is smallest — addressed by more Fourier features and longer training (50K epochs)

Training Convergence

Training Convergence



All runs show the Adam → L-BFGS transition (sharp loss drop at phase boundary). Higher ka requires progressively more compute.

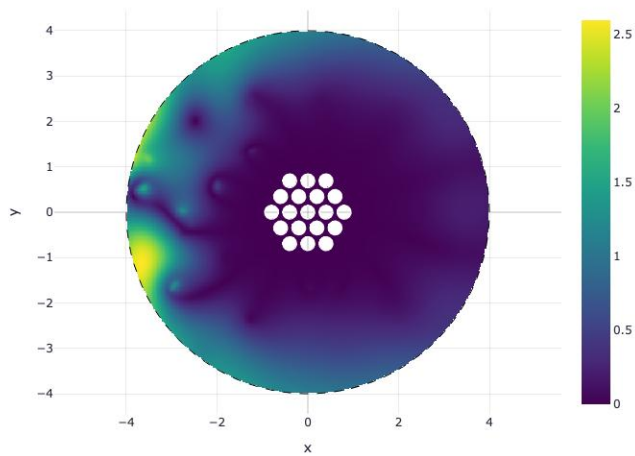
Scaling Across ka

	$KA \leq \pi$	$KA = 2\pi$	$KA = 3\pi$
Neurons	256	384	512
Fourier feat.	64	96	128
Adam epochs	10K	50K	50K
Wall time	12–17 min	259 min	10+ hrs
L2 error	2–8%	2.0%	68%

$ka=3\pi$ loss drops 5 orders of magnitude but L2 only halves — the architecture cannot resolve ~3 wavelengths per radius. Needs multi-scale decomposition or adaptive methods, not just more compute.

Extension: Honeycomb Acoustic Shield

Honeycomb PINN — magnitude_total — ka=2.00



19-circle honeycomb cluster at $ka = 2$, trained as a pure PINN with no analytic reference solution.

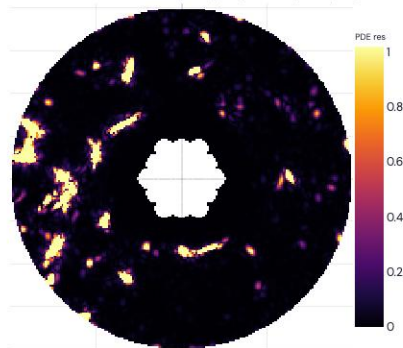
LOSS	VALUE
PDE	$4.92\text{e-}6$
BC (19)	$2.00\text{e-}7$
ABC	$1.41\text{e-}7$
Total	$9.06\text{e-}6$

Shielding: $|\phi_{\text{total}}| < 0.003$ inside cluster.

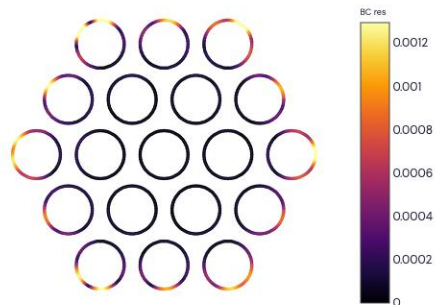
Honeycomb: Residual Diagnostics

Honeycomb Residual Diagnostics ($ka=2$)

PDE Residual $|\nabla^2 \phi_s + k^2 \phi_s|$



Neumann BC Residual (19 surfaces)



PDE residual (left panel): finite-difference Laplacian, 200×200 grid. Near-zero in inter-circle gaps. Elevated near boundaries due to steep field gradients — inherent to the physics.

BC residual (right panel): autograd normal derivatives, 100 pts/circle (1,900 total). Neumann condition satisfied to $O(10^{-3})$ across all 19 surfaces.

Key Takeaways

- 1 Multi-scale accuracy.** 2–8% L2 error from $ka=0.5$ to 2π — a $12\times$ range of spatial frequency with a single architecture.
- 2 Physics-aware features.** Fourier features with $\sigma=k$ embed the wavelength directly, defeating spectral bias at the source.
- 3 Symmetry-matched BCs.** Circle domain + BGT2 absorbing conditions match the cylindrical geometry — physics guides the numerics.
- 4 Training budget matters at high ka .** The convergent 2π run used default loss weights but $2.5\times$ more epochs (50K vs 20K) — patience, not loss rebalancing, drove 49%→2%.

Limitations — $ka=3\pi$

A 512n/6L/128ff run (50K Adam + L-BFGS, 10+ hours) plateaued at **68% L2 error** despite loss dropping 5 orders of magnitude. The loss landscape is being minimized but the field is not resolved — more compute alone won't fix this.

What Would Help

- Multi-scale or domain decomposition
- Residual-adaptive sampling (RAD)
- Curriculum training; low→high ka
- Inverse problems from far-field data

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Code: github.com/carlm451/helmholtzscatteringpinn